

AUTONOMOUS NAVIGATION AND LOCALIZATION FOR
UNMANNED AERIAL VEHICLES IN GPS-DENIED
ENVIRONMENTS: A SYSTEMATIC LITERATURE REVIEW

Version 1 manuscript (V1; N = 171)

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Abstract

GPS/GNSS denial removes a primary source of absolute position and exposes the limits of autonomous aerial navigation. This systematic literature review synthesized methods for localization and navigation of unmanned aerial vehicles (UAVs) in GPS-denied environments. The Version-1 full-text corpus comprised 171 papers (26.9% coverage of the 636-record included set); 465 included records were not retrieved. Eligibility required a UAV-relevant platform, GPS/GNSS denial as the primary contribution, quantitative experimental or simulated results, peer review, publication from 2010-01-01 through 2026-06-30, and English text. A CSV-driven extraction and 0–10 appraisal rubric were audited against the Version-1 PDF set. Quality tiers were Q-high 38/171 (22.2%), Q-medium 84/171 (49.1%), and Q-low 49/171 (28.7%); citation tiers were Core 35/171 (20.5%), Important 87/171 (50.9%), and Peripheral 49/171 (28.7%). Experiments were Real_World 22/171 (12.9%), Simulation 78/171 (45.6%), and Both 71/171 (41.5%). The synthesis indicates a progression from visual odometry and SLAM toward visual-inertial, LiDAR-inertial, radio, and multi-sensor approaches, while real-world and adversarial validation remain uneven. Findings are limited by partial full-text coverage and machine-assisted extraction.

Keywords: UAV; GPS-denied; GNSS-denied; visual-inertial odometry; LiDAR SLAM; sensor fusion; systematic literature review; electronic warfare; localization.

I. Introduction

A. Operational motivation: UAV autonomy in indoor, urban, underground, forest, maritime, and adversarial/electronic-warfare (EW) settings cannot assume continuous GNSS. The review therefore treats denial, degradation, or jamming as a core operating condition rather than a peripheral robustness test.

B. Historical evolution: The V1 corpus spans 2010–2026. The evidence narrative moves from indoor visual odometry and early SLAM, through visual-inertial odometry (VIO), to LiDAR-inertial fusion, radio and cooperative localization, and recent learning- and geo-localization approaches. This is a qualitative evolution statement; no unreported performance values are inferred.

C. Gap and contributions: Prior reviews often combine general navigation with GPS-denied use cases or do not expose full-text coverage. This review contributes: (1) an explicit primary-contribution eligibility test; (2) a reproducible CSV schema and 0–10 quality rubric; (3) a versioned V1 corpus with audit traceability; and (4) separate accounting for real, simulated, and mixed validation.

D. Organization: Section II summarizes related work. Section III specifies protocol and methods. Section IV reports V1 results. Sections V–VII discuss implications, limitations, and conclusions. Appendices and the supplementary package provide reproducibility records.

II. Related Work

The literature contains several recurring method families: visual odometry and VIO; visual, LiDAR, and LiDAR-inertial SLAM; inertial and filter-based navigation; radio/UWB and cooperative localization; map- or geo-localization; and learning-assisted estimators. These families are not mutually exclusive in the extracted records. The present review differs by requiring GPS/GNSS denial to be the primary contribution and by preserving UNKNOWN or NOT_REPORTED values rather than imputing metrics.

III. Review Protocol and Methods

A. Research questions and PICOC: RQ1 asks which methods were proposed during 2010–2026. RQ2 asks which sensor modalities and fusion strategies were used and how prevalence changed. RQ3 asks where methods were validated and the real/simulation ratio. RQ4 asks which open challenges remain, especially in adversarial/EW settings.

The PICOC population was fixed-wing, rotary-wing, and hybrid UAVs; the intervention was localization/navigation without GPS/GNSS; comparisons were method families and time periods; outcomes included accuracy, drift, computation, validation status, and sensor requirements; and context covered indoor, urban, underground, forest, adversarial/EW, and maritime environments.

B. Information sources and search: IEEE Xplore and Scopus were the named databases. The search window was 2010-01-01 through 2026-06-30. Search strings, source logs, and retrieval status are preserved in the S1–S10 supplementary package. No PDFs were downloaded for this manuscript-generation task.

C. Eligibility and screening: All I1–I6 had to hold: UAV platform or explicitly transferable UAV result; GPS/GNSS-denied navigation/localization as the primary contribution; quantitative localization/navigation results; peer-reviewed journal or conference publication; in-window publication; and English language. Any E1–E8 condition excluded a record: theoretical-only; incidental denial; GPS-augmented; out-of-scope subject; non-UAV without transferability; non-peer-reviewed or insufficiently retrievable; outside the window; or non-English. The I2 primary-contribution rule was applied strictly.

D. Deduplication and corpus versioning: Deduplication used DOI-exact matching and title fuzzy matching at the protocol threshold. The manuscript is V1 and uses only the 171-row extracted corpus. The planned target corpus is not substituted into any V1 result. Counts in this document were reconciled to 02_data_processed/extracted_master_v2.csv and 06_analysis/audit/V1_audit_20260917T124255Z.json.

E. Extraction and quality appraisal: The extraction schema includes bibliographic fields, platform, sensors, method category, environment, experiment type, metrics, application, multi-agent status, notes, quality dimensions, quality tier, and citation tier. QA uses rigor (0–4), reporting (0–3), baseline (0–2), and reproducibility (0–1), summed to 0–10. Q-high is 8–10, Q-medium 5–7, and Q-low 0–4. Citation tiers are Core, Important, and Peripheral. Multi-label sensor and environment fields are not mutually exclusive.

F. Audit and reproducibility: The audit reported 170 PASS and 1 PASS_EXCEPTION. The exception was REC_1137: equation-dense text produced a 52.5% short-token ratio; manual review found no genuine extraction failure. The reproducibility package identifies every source path, schema, quality row, audit status, and seed-42 spot-check row.

IV. Results

A. PRISMA flow and corpus coverage: The review flow yielded 636 stage-1 included records, of which 465 were not retrieved for V1 and 171 full texts were extracted. Coverage was $171/636 = 26.9\%$.

B. Quality and citation tiers: Q-high comprised 38/171 (22.2%), Q-medium 84/171 (49.1%), and Q-low 49/171 (28.7%). Citation tiers were Core 35/171 (20.5%), Important 87/171 (50.9%), and Peripheral 49/171 (28.7%). These are categorical counts from the authoritative CSV, not estimates beyond V1.

C. Experiment type and validation: Real_World validation occurred in 22/171 (12.9%), Simulation in 78/171 (45.6%), and Both in 71/171 (41.5%). Mixed validation was common enough to warrant a separate category; it must not be collapsed into either real or simulation.

D. Methods, sensors, environments, and applications: Method, sensor, environment, platform, and application distributions are provided as CSV-derived tables. Because sensor and environment fields are multi-label, their category totals can exceed 171 and should not be interpreted as mutually exclusive study counts. Metrics reported in the source include ATE/RMSE, drift, success rate, and computational latency; values marked NOT_REPORTED remain missing.

E. Audit result: All 171 extracted records had an audit disposition: 170 PASS and 1 PASS_EXCEPTION. The exception was manually reviewed as an extraction artefact rather than a genuine failure. A separate 20-paper human validation remains a required submission task and is not represented as completed here.

V. Discussion

A. RQ1: method landscape: The V1 record set supports a heterogeneous method landscape rather than a single dominant estimator. Visual-inertial and SLAM families coexist with LiDAR-inertial fusion, filter-based methods, radio/UWB, cooperative methods, and map-based approaches. Method labels should therefore be read as primary categories assigned by the extraction schema.

B. RQ2: sensors and fusion: The corpus emphasizes combinations of IMU with cameras and/or LiDAR, while radio and other modalities appear in narrower roles. Fusion is attractive because it can trade visual observability, geometric structure, and inertial continuity, but sensor availability and computation constrain deployment.

C. RQ3: environments and sim-to-real: Indoor and urban scenarios recur, with underground, forest, maritime, and adversarial/EW settings represented unevenly. The 78 simulation-only and 71 mixed records show substantial dependence on simulated or hybrid evidence; the 22 real-world records provide the direct operational subset.

D. RQ4: open challenges and EW: Open challenges include resilience to jamming and spoofing, degraded visual texture, dynamic scenes, map uncertainty, calibration drift, compute and power limits, and reproducible adversarial benchmarks. EW claims should distinguish denial, degradation, jamming, spoofing, and general GNSS absence.

E. Implications: A credible deployment claim should state sensors, environment, baseline, metrics, and failure modes together. The quality rubric and audit fields make those reporting gaps visible, but they do not establish causal superiority between method families.

VI. Limitations

First, V1 covers 171/636 included records (26.9%); 465 records were not retrieved. Second, extraction was machine-assisted and keyword/regex-oriented, so categorical labels require human validation. Third, one audit row required a documented PASS_EXCEPTION. Fourth, searches were limited to IEEE Xplore and Scopus and the English-language window. Fifth, multi-label categories inflate marginal totals and do not represent independent studies. Sixth, NOT_REPORTED metrics were not imputed, and no ATE or accuracy value is invented.

VII. Conclusion

The V1 evidence base shows a broad, evolving field of GPS-denied UAV navigation, with visual-inertial, SLAM, LiDAR-inertial, cooperative, radio, and learning-assisted families. Validation remains split between simulation and mixed studies, with fewer real-world records. The V1 findings are suitable as a traceable working manuscript, but coverage and human validation gates must be addressed before submission.

Data Availability

The analysis artifacts use relative repository paths. The authoritative extracted data are 02_data_processed/extracted_master_v2.csv; audit evidence is 06_analysis/audit/V1_audit_20260917T124255Z.json and 06_analysis/audit/MANUAL_REVIEW.md.

Conflicts and Funding

No funding or competing-interest statement was provided in the governing files; these fields require author completion before submission.

References

- [1] Page MJ et al., “The PRISMA 2020 statement,” *BMJ*, 2021, Art. no. n71.
- [2] Rethlefsen ML et al., “PRISMA-S,” *Syst. Rev.*, 2021, Art. no. 39.
- [3] Source papers are listed in the repository CSV and reference package; DOI-level bibliography formatting is a submission gate.

Appendix A. PRISMA 2020 checklist (27 items)

Item	Requirement	Location/evidence	Status
1	Title identifies systematic review	Title	PASS
2	Structured abstract	Abstract	PASS
3	Rationale	I–II	PASS
4	Objectives/RQs	III-A	PASS
5	Eligibility criteria	III-C	PASS
6	Information sources	III-B	PASS
7	Search strategy	Supplement S2	PASS
8	Selection process	III-C	PASS
9	Data collection process	III-E	PASS
10	Data items	III-E	PASS
11	Study risk-of-bias/appraisal	III-E	PASS
12	Effect measures	III-E/IV-D	PASS
13	Synthesis methods	IV and tables	PASS
14	Reporting-bias assessment	Supplement S8; author completion	PENDING
15	Certainty assessment	Supplement S8; author completion	PENDING
16	Study selection results	IV-A	PASS
17	Study characteristics	Tables II–VII	PASS
18	Risk-of-bias results	IV-B	PASS
19	Results of individual studies	Quality table	PASS
20	Results of syntheses	IV	PASS
21	Reporting biases	Supplement S8	PENDING
22	Certainty of evidence	Supplement S8	PENDING
23	Discussion interpretation	V	PASS
24	Limitations	VI	PASS
25	Conclusions	VII	PASS
26	Registration/protocol	Data availability and protocol files	PASS
27	Support/funding/conflicts	Conflicts and Funding	PENDING

Appendix B. Search strings

Use the preserved IEEE Xplore and Scopus query exports in 08_docs/SUPPLEMENTARY_V1.md S2. Any final database-specific syntax, date, and last-search timestamp must be completed by the authors.

Appendix C. Quality-control audit

Check	Status
V1 counts reconciled to CSV/audit	PASS
Coverage $171/636 = 26.9\%$	PASS
Legacy values absent from new artifacts	PASS
Multi-label captions identify non-mutual exclusivity	PASS
ATE/accuracy not invented	PASS
Audit disposition 170 PASS + 1 PASS_EXCEPTION	PASS
Manual review exception documented	PASS
20-paper human validation	PENDING
Figures 1–9 listed	PASS
PRISMA 27-item checklist present	PASS
PRISMA-S fields present	PASS
V1/V2 separation	PASS

Appendix D. Version log

V1 uses the 171 full-text records available in the current working corpus. The asset expansion is documented separately and is not in V1 results.